FLUTTER OFFICIAL LEARNING PATH

A. MODULE 1 – Introduction to Flutter & Environment Setup

(Session 1–5 | CLO1, CLO6)

🎯 **Mục đích module**

- Hiểu Flutter là gì, kiến trúc, widget-based UI
- Cài đặt môi trường, làm quen tooling, hot reload

📚 **Ánh xạ session → Flutter Docs**

Session	Nội dung	Flutter Docs
1	Flutter overview	https://docs.flutter.dev/get-started/flutter-for
1–2	Cài Flutter SDK, IDE	https://docs.flutter.dev/get-started/install
2	Project structure	https://docs.flutter.dev/development/tools/sdk/overview
3	Hot reload, DevTools	https://docs.flutter.dev/tools/hot-reload
3	Widget tree	https://docs.flutter.dev/development/ui/widgets-intro
4	Scaffold, MaterialApp	https://docs.flutter.dev/development/ui/widgets/material
5	Tổng kết & best practices	https://docs.flutter.dev/resources/architectural-overview

🧪 **Codelabs khuyến nghị**

- https://codelabs.developers.google.com/flutter-intro
- https://codelabs.developers.google.com/codelabs/flutter-codelab-first

B. MODULE 2 – Dart Essentials

(Session 6–10 | CLO1, CLO2, CLO4)

🎯 **Mục đích module**

- Nắm vững Dart syntax, OOP, async nền tảng
- Chuẩn bị cho Future / Stream / Flutter logic

📚 **Ánh xạ Flutter & Dart Docs**

Session	Nội dung	Link
6–9	Dart cơ bản (types, functions, classes, collections)	https://dart.dev/guides/language/language-tour
10	Future & async/await	https://dart.dev/codelabs/async-await

🧪 **Codelabs**

- https://codelabs.developers.google.com/dart-cheatsheet
- https://dart.dev/codelabs/async-await

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C. MODULE 3 – OOP & Asynchronous Programming

(Session 11–15 | CLO1, CLO2, CLO3)

🎯 **Mục đích**

- Thành thạo Future, Stream
- Áp dụng reactive programming trong Flutter

📚 **Flutter Docs**

Nội dung	Link
Futures	https://docs.flutter.dev/cookbook/networking/fetch-data
Streams	https://dart.dev/tutorials/language/streams
StreamBuilder	https://docs.flutter.dev/development/data-and-backend/state-mgmt/simple
Async patterns	https://docs.flutter.dev/resources/architectural-overview

🧪 **Codelabs**

- https://codelabs.developers.google.com/codelabs/flutter-firebase
- https://codelabs.developers.google.com/flutter-streams

D. MODULE 4 – UI Fundamentals

(Session 17–20 | CLO1, CLO3)

🎯 **Mục đích**

- Xây UI chuẩn Material
- Làm việc với ListView, widgets tương tác

📚 **Flutter Docs**

Nội dung	Link
Widgets & layout	https://docs.flutter.dev/development/ui/widgets
Lists	https://docs.flutter.dev/cookbook/lists
Input widgets	https://docs.flutter.dev/development/ui/widgets/input
Theming	https://docs.flutter.dev/cookbook/design/themes

🧪 **Codelabs**

- https://codelabs.developers.google.com/flutter-layout
- https://codelabs.developers.google.com/flutter-material

E. MODULE 5 – Navigation & State Management

(Session 21–25 | CLO1, CLO5, CLO7)

🎯 **Mục đích**

- Điều hướng đa màn hình
- Hiểu Navigator 1.0 & 2.0, deep link

📚 **Flutter Docs**

Nội dung	Link
Navigator 1.0	https://docs.flutter.dev/development/ui/navigation
Named routes	https://docs.flutter.dev/cookbook/navigation/named-routes
Navigator 2.0	https://docs.flutter.dev/development/ui/navigation/router
Deep linking	https://docs.flutter.dev/development/ui/navigation/deep-linking

🧪 **Codelabs**

- https://codelabs.developers.google.com/flutter-navigation

F. MODULE 6 – Layout & Responsive Design

(Session 26–30 | CLO2, CLO5, CLO7)

🎯 **Mục đích**

- Thiết kế UI responsive cho mobile/tablet
- Hiểu constraint & adaptive UI

📚 **Flutter Docs**

Nội dung	Link
Row / Column / Flex	https://docs.flutter.dev/development/ui/layout
Grid / Stack	https://docs.flutter.dev/cookbook/layout/grid-lists
MediaQuery	https://docs.flutter.dev/cookbook/design/orientation
Responsive design	https://docs.flutter.dev/ui/layout/adaptive-responsive

🧪 **Codelabs**

- https://codelabs.developers.google.com/flutter-adaptive-app

G. MODULE 7 – Forms & Validation

(Session 32–35 | CLO1, CLO7)

🎯 **Mục đích**

- Xử lý form, validation, UX input

📚 **Flutter Docs**

Nội dung	Link
Forms	https://docs.flutter.dev/cookbook/forms
Validation	https://docs.flutter.dev/cookbook/forms/validation
Focus & keyboard	https://docs.flutter.dev/cookbook/forms/focus

H. MODULE 8 – Networking & REST API

(Session 36–40 | CLO2, CLO6)

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🎯 **Mục đích**

- Gọi API, parse JSON, xử lý loading/error

📚 **Flutter Docs**

Nội dung	Link
HTTP	https://docs.flutter.dev/cookbook/networking/fetch-data
JSON parsing	https://docs.flutter.dev/cookbook/networking/background-parsing
FutureBuilder	https://docs.flutter.dev/cookbook/networking/fetch-data

🧪 **Codelabs**

- https://codelabs.developers.google.com/flutter-rest-api

I. MODULE 9 – Local Storage & Persistence

(Session 41–45 | CLO3, CLO5)

🎯 **Mục đích**

- Lưu dữ liệu local, offline-first

📚 **Flutter Docs**

Nội dung	Link
SharedPreferences	https://docs.flutter.dev/cookbook/persistence/key-value
SQLite	https://docs.flutter.dev/cookbook/persistence/sqlite

J. 🔷 MODULE 10 – Authentication & Session

(Session 46–50 | CLO1, CLO2)

📚 **Flutter Docs**

- https://docs.flutter.dev/development/data-and-backend/state-mgmt
- https://docs.flutter.dev/cookbook/navigation/navigation-basics

🧪 **Codelabs**

- https://codelabs.developers.google.com/flutter-firebase-auth

K. 🔷 MODULE 11–12 – Testing, Build & Release

(Session 51–57)

📚 **Flutter Docs**

Nội dung	Link
Unit & widget test	https://docs.flutter.dev/testing
Debugging	https://docs.flutter.dev/tools/devtools
Build & release	https://docs.flutter.dev/deployment